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BEHAVIORAL AND PITUITARY – ADRENAL RESPONSES DURING A PROLONGED SEPARATION PERIOD IN INFANT RHESUS MACAQUES

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SUMMARY

(1) Separation of infant rhesus macaques from their mothers for 2 weeks induced brief plasma cortisol elevations which returned to near pre-separation levels within 24 hr. (2) Changes in locomotion and distress (whoo) calls paralleled the changes within 24 hr after separation in pituitary – adrenal activity, thus indicating a short-term biobehavioral response to separation in the infant rhesus macaque. (3) Separated infants were behaviorally agitated throughout the separation period, and some behaviors indicative of the "despair" response were infrequently observed. (4) Individual differences in pituitary – adrenal response to separation and reunion were related to aspects of the mother – infant relationship prior to separation.

Key Words – Pituitary – adrenal system; attachment; separation; rhesus macaques.

INTRODUCTION

SEPARATING the primate infant from its mother is viewed as a highly stressful procedure (Seay, Hansen & Harlow, 1962; Kaufman, 1973). The infant shows an initial "protest" response, consisting of increased locomotion, distress calls, and intense agitation. Unless the infant is reunited with the mother, this protest reaction may be followed in several days by a "despair" response, involving hypolocomotion, a collapsed posture, few vocalizations and a withdrawn state (Kaufman, 1973). The intensity and duration of each phase of response may vary as a function of the separation environment, species studied, or the infant's prior experiences (Mineka & Suomi, 1978). In general, variations in the infant's behavioral response to separation have been used to infer the extent of distress resulting from the separation (Hinde & Davies, 1972; Chappell & Meier, 1975; Mineka & Suomi, 1978). The activity of the pituitary – adrenal system has been shown to be a sensitive and reliable index of physiological arousal (Hennessy & Levine, 1979). Recently, the activity of this endocrine system has been used to examine the infant's response to brief separations (Coe, Mendoza, Smotherman & Levine, 1978; Mendoza, Smotherman, Miner, Kaplan & Levine, 1978; Smotherman, Hunt, McGinnis & Levine, 1979; Hennessy, Kaplan, Mendoza, Lowe & Levine, 1979). Separations of 30 min in squirrel monkeys and

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3 hr in rhesus monkeys have been examined using adrenocortical as well as behavioral measures. The results of these studies raise questions about conclusions regarding arousal based solely on the infant's behavioral responses to separation. In most cases separation from the mother has produced behavioral responses indicative of distress along with significant elevations in plasma corticoids. However, dissociations between plasma cortisol and behavioral responses have been observed. For example, mother-reared squirrel monkey infants who received maternal care from a late pregnant female ("aunt") during a half-hour separation showed no agitation or protest behavior. Nonetheless, they showed elevations in plasma cortisol which were as great as those observed when they were housed alone during separation (Coe *et al.*, 1978). Conversely, surrogate-reared squirrel monkeys failed to show elevations in plasma corticoids when their surrogate was removed, even though they exhibited behavioral responses (i.e. increased vocalizations), which have been traditionally interpreted as indicative of an agitated state (Hennessy *et al.*, 1979). Thus, behavioral and physiological measures can provide different, sometimes apparently contradictory information about the infant's state.

The following study was conducted to examine the antecedents of behavioral and pituitary – adrenal responses and the relationship between these responses during a more prolonged (12-day) separation in infant rhesus monkeys. To obtain measures of pituitary – adrenal activity, both the infant and mother were removed from their social groups and were housed individually during separation. Hinde & Davies (1972) and Hinde & McGinnis (1977) have reported that separating the infant by removal from its social group increases the duration of agitated protest behavior, and delays the onset of despair reactions. We were therefore interested in determining whether plasma corticoids would remain elevated under conditions producing prolonged behavioral agitation. A second purpose of the study was to examine the relationship between aspects of the mother – infant relationship prior to separation, and the infant's behavioral and pituitary – adrenal responses to separation and reunion. Reunion following separation sometimes appears to be stressful for both mother and infant (Hinde & Davies, 1972). Hinde & McGinnis (1977) found that tension between the mother and infant rhesus monkeys prior to separation predicted greater behavioral distress upon reunion. Bowlby (1973) has also argued that the quality of the mother – infant relationship should influence the response to both separation and reunion. Thus, the relationship between behavioral and pituitary – adrenal responses during separation, as well as the relationship between an index of mother – infant interaction and the infant's responses to separation and reunion were investigated.

METHODS

Subjects and conditions

The subjects were 9 rhesus monkey infants (7 males, 2 females) and their mothers. The infants were between 5 and 11 months of age ($\bar{X}$ = 7.2 months). Five of the mother – infant pairs were separated during the course of the study, while 4 served as nonseparated controls. The subjects were from 2 social groups, each consisting of 1 adult male, 5 mother – infant pairs (1 infant was not used in the study because of illness), several adult females without infants, and several juvenile females born into the colony during previous birth seasons. The groups lived in 2 large outdoor enclosures (5 × 10 × 3 m), each adjoined by a smaller enclosed room (3 × 5 × 3 m). The groups had been established several years prior to this study. The study was conducted in two complete replications during the months of November through January. The first replication involved 2 separated and 2 nonseparated pairs, while the second replication involved 3 separated and 2 nonseparated pairs.

Procedure

All subjects were captured out of their groups and rehoused with their mothers in individual-size (64 × 74 × 30 cm) cages for 7 days. They were given 3 days to adapt to these conditions prior to obtaining pre-separation measures of plasma cortisol. The subjects were located in 2 rooms in separate buildings. During adaptation, one room contained the mother – infant pairs who would later be separated, while the other room contained the mothers and infants who would not be separated. The separation manipulation took place on the 7th day following capture at 15:00 hrs. Separation involved removing the mothers and housing them in individual cages in the room containing the nonseparated controls. To separate the mother and infant, the mother was first restrained using a squeeze apparatus. The infant was then taken from the cage, the mother was captured and removed, and the infant was immediately replaced in the cage it had shared with its mother during the adaptation phase. To control for the separation procedure, nonseparated infants were also removed from their cages, handled briefly (30 sec) and then reunited with their mothers.

The next 12 days constituted a period of total separation for the experimental subjects. Separated infants could see and hear other separated infants, and those housed in adjacent cages could achieve limited tactile contact. However, during this period no form of contact between the separated infant and its mother was possible. Following this period of total separation, the separated infants served as stimuli in a study examining the specificity of the mother's responses to her infant (Gonzalez, Gunnar & Levine, 1981). During the course of that study, all infants experienced 8 additional days of separation, during which they had visual and auditory access once to their own mother, and once to another separated mother for 1 hr. Reunion took place at 16:00 hrs on the last day of the second study. Each separated infant was placed with its mother in the cage the mother had occupied during the separation period. Thus, the separated infant, rather than the mother, was relocated in order to reduce the possibility that the mother would be agitated following reunion as a function of her being captured and rehoused. A blood sample was obtained 24 hr after reunion, then all subjects were released back into their social groups.

Plasma cortisol

Measures of plasma cortisol were obtained from blood samples collected on days 3 and 5 of the adaptation phase, on days 1 (½ hr and 3 hr after separation), 2, 3, 6, 9 and 12 of the total separation phase, and 24 hr following reunion. All samples were taken between 15:00 and 16:00 hrs, with the exception of the sample obtained 3 hr after separation, which was taken at 18:00 hrs. All samples (0.5 ml) were obtained via saphenous venipuncture using heparinized tuberculin syringes. Infants with their mothers were sampled by restraining the mother using a squeeze-cage apparatus, and withdrawing the infant's leg through an opening at the bottom of the cage. Separated infants were quickly removed from their cages and held for sampling. The time it took to sample any given infant ranged from 1 to 5 min. The order in which subjects were sampled in each room varied randomly. Cortisol was assayed using antiserum No. F21-53 from Endocrine Sciences, Tarzana, CA as described by Klemm & Gupta (1975). The method used in this laboratory has also been described in detail by Smotherman *et al.* (1979).

Behavioral observations

Mothers and infants were observed in their social groups for the 10 days immediately preceding capture and rehousing. Each pair was observed daily for 5 min in the morning (09:00 – 10:00 hrs.) and 5 min in the afternoon (15:00 – 16:00 hrs) by 2 observers seated behind a blind. Mother – infant contact (clinging vs passive) and proximity (within 12 cm) were recorded on the signal every 20-sec while each infant or mother approach and leave was noted. Percent agreement estimates of inter-observer reliability were obtained for each replication group using ten 5-min observations. Reliability for each measure was greater than 75%. Two scores were then computed for each mother – infant pair. One score reflected the degree of contact between mother and infant. This score was computed by multiplying each instance of clinging by 3, passive contact by 2, and proximity by 1, so as to give the most active form of contact the most weight. One such score was obtained for each day, and the average score over the 10 days of observations was taken. The second score was the Hinde index (Hinde & Spencer-Booth, 1967) of the infant's role in maintaining proximity (% Approach – % Leave). This score was computed by dividing the number of infant approaches during each day's observations by the number of both infant and mother approaches, and subtracting from this score the number of infant leaves divided by the number of both infant and mother leaves. One score over the 10 days of observations was then computed by taking the average of all days. The same observational procedures were followed, and measures of the degree of contact and % Approach – % Leave were obtained for the 5 days following return to the social group. On each of these 5 days, however, each mother – infant pair was observed for 20 min rather than 10 min (10 min in the morning, 10 min in the afternoon) to increase the likelihood of discerning changes in the relationship.

During the total separation period each infant (separated and nonseparated) was observed for 30 min per day (15 min between 09:00 and 10:00 hrs and between 15:00 and 16:00 hrs) by an observer seated behind a screen in

front of the infant's cage. The observations were made using check sheets divided into fifteen 20-sec intervals for each 5 min of observation. The behaviors recorded were (1) infant state (see Table I) which was recorded on the signal every 20 sec; (2) 3 types of infant posture (sitting/standing upright, hunched-over or curled-up into a ball) which were recorded on the signal; (3) 2 types of vocalizations (whoo calls and screeches) which were recorded as occurrence/non-occurrence during each coding interval; (4) each instance of movement, which was defined as each change of location of 5 in. with a 5-sec pause constituting the division between two instances of movement. The room housing the separated infants was proximal to other animal rooms and caretaker facilities. This meant that the ambient noise level varied between observation periods, which in turn affected the behaviors observed. To stabilize the behavioral data, scores for each of the measures were averaged into six 2-day blocks for the 12 days of observations. All analyses were conducted using these averaged scores.

TABLE I. DEFINITIONS OF BEHAVIORAL STATES

State	Definition
Alert	The infant is awake, eyes are focused, muscle tone is firm
Agitated	Muscle tone is rigid, movements are rapid, may be exhibiting sterotypes, vigilant, hyperresponsive to noise
Quiet – relaxed	Muscle tone firm, breathing rhythmic and deep, eyes glassy, lids half-closed
Quiet – withdrawn	Eyes open, staring – glazed, tone flaccid, unresponsive to noise, may be exhibiting digit sucking or other forms of self-stimulation, movements slow
Sleep	Eyes closed for at least 1 min (3 coding intervals), breathing rhythmic, muscle tone relaxed

TABLE II. PERCENTAGE OF TIME IN EACH STATE[a]

	State				
Group	Sleep	Quiet relaxed	Alert	Agitated	Quiet withdrawn
Nonseparated	5	12	80	3	0
Separated	0	1	66	32	1

[a]Average of all 12 days.

RESULTS

Table II shows the average percentage of time spent in each of the 5 behavioral states by nonseparated and separated infants. Over all 12 days, nonseparated infants averaged 80% of the time in an alert, non-agitated state. A quiet, relaxed state was the second most frequently observed state for these infants. In contrast, separated infants, although alert and non-agitated for over half of the observation intervals, were agitated significantly more of the time than were nonseparated infants ($t(7) = 2.68$, $p < 0.05$). The despair response was never the modal response pattern for any of the infants during separation.

However, all infants showed some components of this response. Beginning in the second week of separations, 3 of the 5 infants were observed in a quiet, withdrawn state during 1 – 10% of the coding intervals during a given observation period; but the frequency of quiet, withdrawn state fluctuated greatly from day to day. None of the infants ever evinced the extreme despair posture (curled up into a ball), but all showed the hunched-over posture at some point during separation. This posture was never observed for more than an average of 10% of the coding intervals for any 2-day block of observations.

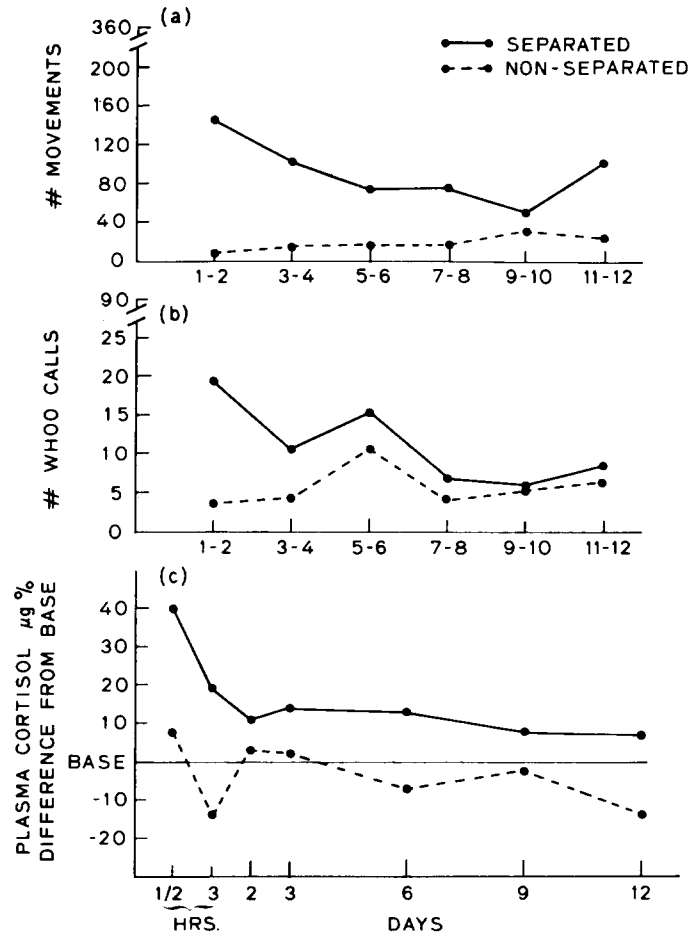


FIG. 1. Number of movements (a), number of whoo calls (b), and plasma cortisol (µg%) difference from pre-separation levels (base), (c) for ½ hr, 3 hr, and days 2, 3, 6, 9 and 12 after separation.

Agitated state was recorded so rarely for infants who were with their mothers that changes in agitated state during the separation period could not be examined for separated and nonseparated infants in the same analysis. A one-way ANOVA was conducted on the separated infants' data and showed that, although there appeared to be some reduction in agitation over the 12 days, variability was great and the reduction was not significant

[$F(5,20) = 0.44$]. Movement and whoo calling occurred frequently enough in nonseparated infants for the 2 groups to be compared directly, using a 2(groups) × 6(time blocks) repeated measures ANOVA (see Fig. 1a and 1b). Both measures yielded significant Groups × Time Blocks interactions, [$F(5,35) = 3.79$, $p < 0.01$ (movement); $F(5,35) = 3.09$, $p < 0.05$ (whoo calls)]. Tests for simple main effects and subsequent Newman–Keuls analyses (Winer, 1971) indicated that neither movement nor whoo calling varied over the 6 time blocks for the nonseparated infants. For the separated infants both movement and whoo calling were elevated during Block 1 (days 1–2), but by Block 2 (days 3–4) had returned to levels which were not significantly different from those observed in nonseparated infants.

Plasma cortisol data for the infants were analyzed using a 2(group) × 8(time) repeated measures ANOVA (see Fig. 1c). The average of the 3rd and 5th day of adaptation was considered a measure of pre-separation (basal) cortisol level for each infant. A significant group × time interaction was observed [$F(7,49) = 3.44$, $p < 0.01$]. Tests for simple main effects of time and subsequent Newman–Keuls tests showed that cortisol levels for nonseparated infants were never significantly elevated over their pre-separation (basal) level. In contrast, separated infants showed significant elevations ½ hr and 3 hr after separation. By 24 hr after separation cortisol levels were not significantly elevated over adaptation levels. These data are presented graphically in Fig. 1c in terms of differences from pre-separation levels, and in tabular form in Table III in terms of absolute values. To

TABLE III. PLASMA CORTISOL LEVELS DURING ADAPTATION SEPARATION AND REUNION

Groups	Adaptation (X Days 3 and 5)	Separation ½ hr	3 hr	24 hr	Day 3	Day 6	Day 9	Day 12	Reunion
Separated	16.7[a] (.92)	58.3 (8.8)	38.0 (3.9)	29.4 (5.9)	32.3 (5.3)	31.6 (4.2)	26.7 (3.5)	25.8 (7.9)	28.8 (6.4)
Nonseparated	27.5 (4.7)	36.3 (6.3)	14.7 (2.1)	31.8 (2.8)	30.2 (4.6)	21.2 (2.1)	26.7 (4.5)	14.7 (2.4)	25.9 (1.3)

[a]Mean and (standard error of the mean) expressed as μg%.

analyze further the relationship between cortisol reductions and decline in behavioral responses for the separated infants, Spearman rank-order correlations were computed (see Table IV). The results showed that the reduction in whoo calling, but not movement, between Blocks 1 and 2 was directly related to the magnitude of the reduction in plasma cortisol from ½ hr to 24 hr after separation [$\rho(5) = 0.90$, $p < 0.05$ (whoo calling); $\rho(5) = 0.30$, NS (movement)]. The effects of reunion on pituitary–adrenal activity were examined by comparing values obtained 24 hr after reunion with pre-separation values. A 2(group) × 2(pre-separation/reunion) repeated measures ANOVA failed to reach statistical significance. However, as seen in Table IV, 4 of the 5 separated infants had higher reunion than pre-separation levels of plasma cortisol. This was the case for only 1 of the 4 nonseparated infants. Furthermore, for separated infants increases ranged from 1.0 to 27.1 μg%.

TABLE IV. REDUCTION IN PLASMA CORTISOL FROM ½ HR TO 24 HR AFTER SEPARATION AND CHANGES IN BEHAVIORAL RESPONSES

Subject no.[a]	Cortisol (μg%) 24 hr – ½ hr	Whoo calls (BL 2 – BL 1)	Movement (BL 2 – BL 1)
4	−75.7	−12.0	+3.5
3	−36.2	−11.0	+20.8
5	−20.5	−6.0	+70.7
1	−12.3	−1.0	+4.6

[a]Subject numbers correspond to those in Table V. Plasma cortisol data were obtained for only 4 separated infants 3 hr after separation.

Behavioral responses following return to the social group were examined by comparing each subject's average score for "degree of contact" and "% Approach – % Leave" on the 10 days prior to capture with its average score during the 5 days following return to the group. A 2(group) × 2(pre/post) repeated measures ANOVA showed no significant changes in degree of contact following these manipulations for either the separated or the nonseparated mother–infant pairs. A similar analysis conducted on the Hinde % Approach – % Leave index, however, showed that for both separated and nonseparated pairs there was a significant increase in the infant's role in maintaining contact with the mother following return to the social group [$F(1,7) = 38.93, p < 0.001$]. Finally, the Hinde index of % Approach – % Leave was used to examine the effects of "tension" in the mother–infant relationship on both behavioral responses to separation and pituitary–adrenal response to both separation and reunion (see Table V). No significant correlations were observed between the Hinde index and behavioral responses to the mother's removal. In contrast, significant correlations were found between the Hinde index and the pituitary–adrenal responses to both separation [$\rho(5) = -0.90, p < 0.05$] and reunion [$\rho(5) = +0.90, p < 0.05$].

TABLE V. THE RELATIONSHIP BETWEEN A MOTHER–INFANT DYAD'S HINDE INDEX SCORE PRIOR TO SEPARATION AND THE INFANT'S BEHAVIORAL AND PHYSIOLOGICAL RESPONSES TO SEPARATION AND REUNION

			Separation		Reunion	
Separated pairs	% Approach – % Leave (pre-separation)	Whoo calls[a]	Movement	Agitation	Cortisol[b]	Cortisol[c]
1	0.46	11.0	38.5	20.0	+9.3	+27.1
2	0.31	36.5	216.0	77.5	+13.7	+12.7
3	0.29	13.0	75.0	22.5	+26.5	+20.5
4	0.19	13.5	80.0	1.5	+27.3	+1.0
5	0.00	23.5	326.0	88.0	+29.8	−0.7

[a]Levels of whoo calling, movement, and agitated state, Block 1 (days 1–2).
[b]3 hr separation value minus $\bar{X}$ days 3 and 5 of adaptation expressed as μg%.
[c]24 hr reunion value minus $\bar{X}$ days 3 and 5 of adaptation expressed as μg%.

DISCUSSION

Infants housed with their mothers (nonseparated) were in an alert, non-agitated state during most of the study with the next most frequently observed behavioral state being "quiet – relaxed". Plasma cortisol levels for nonseparated infants averaged 25.8 μg%. These levels are slightly higher than basal values reported in other studies (Hill, McCormack & Mason, 1973), but are clearly lower than levels considered indicative of stress. Individual nonseparated infants showed day-to-day fluctuations in plasma cortisol levels, but no consistent changes were observed over days. Indeed, although these infants were captured and handled briefly when the other infants were separated from their mothers, these manipulations did not produce significant elevations in corticoids. Similar findings have been reported for squirrel monkey infants (Coe *et al.*, 1978). It appears that in both species capture and handling does not activate the pituitary – adrenal system if the infant is immediately reunited with its mother. Maternal contact may buffer physiological responses to stressors in infancy (Mendoza *et al.*, 1978). In sum, these rhesus infants were not seemingly disturbed by being removed from their social groups and housed in individual-sized cages, as long as they were with their mothers.

In contrast, separating the infant from its mother produced striking behavioral and physiological responses. Similar to Hinde (Hinde & Davies, 1972; Hinde & McGinnis, 1977) we found that infants housed alone during separation showed a prolonged period of behavioral agitation, which continued throughout the 2-week separation period. Although they were frequently agitated, some components of the "despair" response began to appear in the second week. Despair, however, never characterized the state of any of the infants. Rhesus infants left in their social groups during separation typically show a transition from agitated to depressed behavior by the 2nd or 3rd day of separation. Infants in the present study showed changes in 2 behaviors, whoo calling and movements, between days 1 – 2 and days 3 – 4 of separation. Recently, Rosenblum & Plimpton (in press) identified specific vocalizations and locomotion as behaviors directed towards the reestablishment of maternal contact. These behaviors may show a short-term increase following separation during the period when the probability of relocating the mother is high (e.g. the first day or so). Reductions in these behaviors may signify that the infant has entered a second phase of separation; a phase which may or may not be characterized by an increase in behaviors indicative of "despair" or "depression" (Mineka & Suomi, 1978).

Separation from the mother also producing striking elevations in plasma cortisol. After ½ and 3 hr following separation from their mothers, infant cortisol levels were extremely elevated. By 24 hr after separation, and for the remainder of the separation period, cortisol levels were not significantly higher than pre-separation levels. Given the small sample size ($n = 5$), we are hesitant to conclude that the separated infants had actually returned to pre-separation levels by 24 hr after separation. Indeed, an examination of Fig. 1c indicates that although average cortisol values fluctuated around pre-separation levels for nonseparated infants, for separated infants average levels never fell below average pre-separation levels throughout the separation period. Thus, there is some suggestion that the separated infants had slightly elevated cortisol levels during separation. Clearly, however, there was a significant reduction in pituitary – adrenal activity during the first 24

hr after removal of the mother. Similar changes in other physiological systems (e.g. heart rate and body temperature) have been reported at 24 hr in other laboratories (Reite, Short, Kaufman, Stynes & Pauley, 1978). The existence of an initial short-term response to separation is further supported by correlations between reductions in plasma cortisol levels and reductions in behavioral responses to separation. Infants who showed the greatest reductions in plasma cortisol levels from ½ hr to 24 hr of separation, subsequently showed the greatest reductions in whoo calls. Such biobehavioral relationships highlight the importance of examining changes in specific behaviors during separation, in addition to tracking changes in more global behavioral states.

One of the primary purposes of the study was to examine relationships between aspects of the mother – infant relationship and the infant's responses to separation and reunion. The Hinde index of % Approach – % Leave was used as a measure of tension in the mother – infant relationship surrounding the maintenance of proximity. Values approaching 1.0 indicate that the infant does most of the approaching, while the mother does most of the leaving, while the reverse is indicated by values approaching −1.0. Values close to 0.0 indicate infant and mother engage in about equal amounts of approaches and leaves (Hinde & Spencer-Booth, 1967). No significant relationships were found between this measure and either behavioral or physiological responses for nonseparated infants. In contrast, several significant relationships were observed for separated infants. First, higher % Approach – % Leave scores were negatively related to pituitary – adrenal responses to separation. That is, in pairs with a more "balanced" (closer to 0.0) proximity-maintaining relationship, the infant showed a greater pituitary – adrenal response to separation. One possible explanation of this effect is that the % Approach – % Leave index reflects the infant's expectancy of being able to achieve and maintain contact with the mother, with lower values reflecting higher expectancies. If so, then enforced separation from the mother may have been experienced as a greater violation of expectancy for infants in more "balanced" relationships. Previous research has shown that the pituitary – adrenal system is sensitive to expectancies and violations of expectancies (Hennessy & Levine, 1979). Alternatively, infants who have had greater access to their mothers may have had less experience in dealing with perturbation in the environment without maternal aid and contact. Maternal loss might therefore constitute a greater "threat" for these infants, producing a more marked pituitary – adrenal response. There are not sufficient data to conclusively support either possible explanation. However, these data do indicate that pre-separation aspects of the mother – infant relationship influences the infant's responses to separation. This relationship has not been clearly demonstrated using behavioral measures in response to separation (Ainsworth, Bell & Stayton, 1971), and indeed was not found for the measures of infant behavior in the present study.

Reuniting the infants with their mothers did not produce consistent changes in plasma cortisol levels. Some infants exhibited levels 24 hr after reunion similar to those pre-separation, while others showed elevations of as much as 27 μg%. Elevations in plasma cortisol levels over pre-separation values were positively correlated with pre-separation measures of % Approach – % Leave. The more "imbalanced" the pre-separation relationship (closer to 1.0) the greater the elevations in plasma cortisol following reunion.

This finding is congruent with attachment theory (Bowlby, 1973) which predicts that reunion between mother and infant will be more stressful if there has been tension in the relationship prior to separation. The significance of this finding is enhanced by the fact that reunion cortisol values were obtained 24 hr after reunion, a time by which, following separation, the effects of maternal loss on pituitary–adrenal activity were much abated. In combination with the correlations found during separation, these data indicate that for some infants adapting to maternal loss may be less stressful than adapting to reunion with the mother. In considering this hypothesis, it is important to note that infants in the present study were of an age where they could adapt and survive in the absence of the mother. Results might be very different if younger, less socially and physically competent subjects had been examined. Finally, following return to the social group, both separated and nonseparated pairs showed an increase in the % Approach – % Leave Hinde index relative to values obtained before their removal with their mothers from the group. This probably reflected reduced maternal responsiveness due to the mother's need to reestablish her position in the dominance structure of the group (Hinde & McGinnis, 1977). Since no significant difference was observed between separated infants and nonseparated infants after return to the group, this suggests that the effects of removing and returning mother and infant to their social groups may perhaps obscure the effects of separation *per se* on the relationship.

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